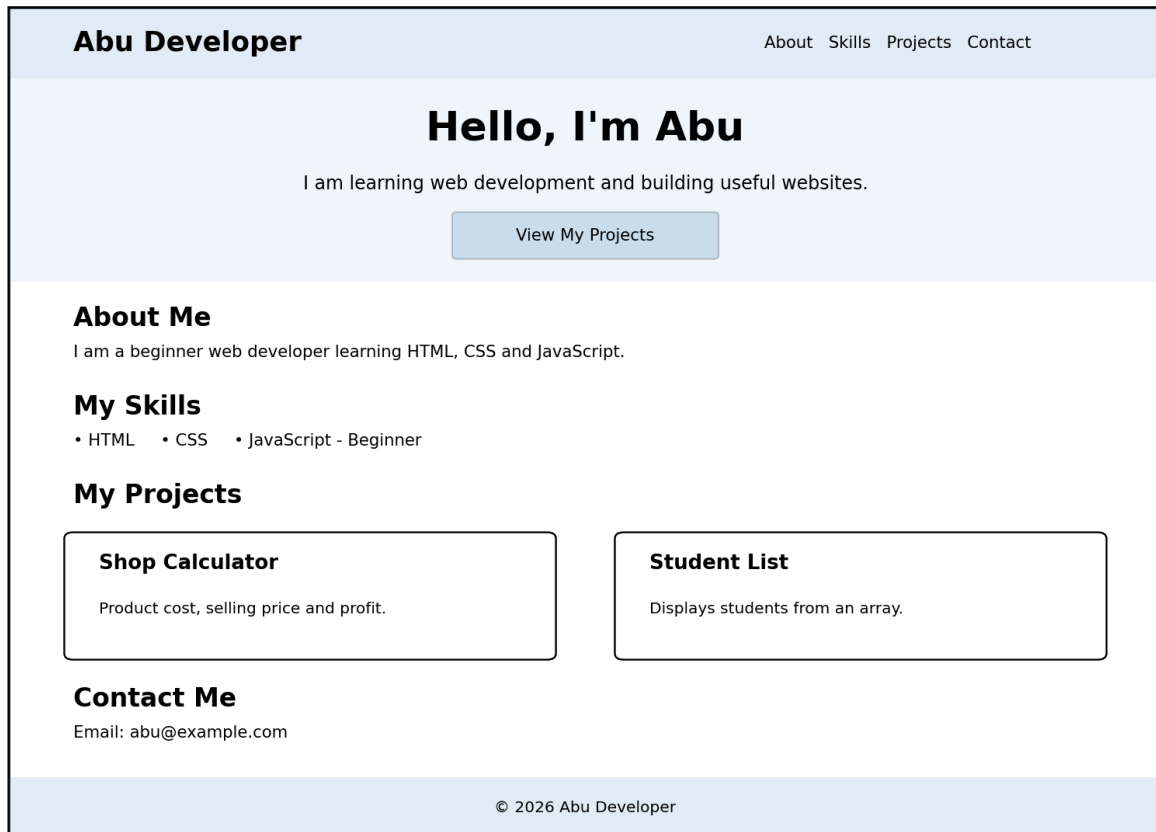


BUILD A SIMPLE PORTFOLIO WEBSITE

Project goal: build a complete one-page personal portfolio while learning what each important line of HTML and CSS does.



Expected design outcome

1. What you will learn

- How HTML creates the structure of a webpage.
- How CSS changes colors, spacing, fonts and layout.
- How to connect an external CSS file to HTML.
- How IDs and classes are used.
- How navigation links jump to sections on the same page.
- How to create reusable project cards.
- How to make a basic responsive layout for smaller screens.

2. Understand HTML and CSS first

Think of a website like a house. HTML is the structure of the house, while CSS is the decoration and arrangement.

Technology	Job	Example
HTML	Structure and content	Headings, paragraphs, images, buttons
CSS	Design and layout	Colors, spacing, borders, positioning
Browser	Displays the result	Chrome, Edge, Firefox

3. Create the project

Create a folder named my-portfolio. Inside it create two files:

```
my-portfolio/  
├── index.html  
└── style.css
```

Important: The file names must match the names used in the tutorial. Save both files in the same folder.

4. Start with the basic HTML document

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Abu Portfolio</title>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  
</body>  
</html>
```

Explanation

<!DOCTYPE html> - Tells the browser that this document uses modern HTML5.

<html lang="en"> - The root element of the webpage. lang="en" says the page language is English.

<head> - Contains information about the page that is mainly used by the browser.

<meta charset="UTF-8"> - Allows the page to correctly display common characters and symbols.

viewport meta tag - Helps the website fit correctly on phones and tablets.

<title> - The text displayed on the browser tab.

<link rel="stylesheet"> - Connects index.html to style.css.

<body> - Contains everything the visitor can see on the webpage.

5. Add the header and navigation

```
<header>
  <h1>Abu Developer</h1>

  <nav>
    <a href="#about">About</a>
    <a href="#skills">Skills</a>
    <a href="#projects">Projects</a>
    <a href="#contact">Contact</a>
  </nav>
</header>
```

The header is the top section of the website. The <nav> element contains navigation links.

How the links work: href="#about" looks for an element with id="about". Clicking the link moves the browser to that section.

6. Add the hero section

```
<section class="hero">
  <h2>Hello, I'm Abu</h2>
  <p>I am learning web development and building useful websites.</p>
  <a class="button" href="#projects">View My Projects</a>
</section>
```

The hero is the first major introduction visitors see. class="hero" gives us a name we can target from CSS. The button is actually an <a> link styled to look like a button.

7. Add About, Skills, Projects and Contact

```
<section id="about">
  <h2>About Me</h2>
  <p>I am a beginner web developer learning HTML, CSS and JavaScript.</p>
</section>

<section id="skills">
  <h2>My Skills</h2>
  <ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript - Beginner</li>
  </ul>
</section>

<section id="projects">
  <h2>My Projects</h2>
```

```

<div class="project-card">
  <h3>Shop Calculator</h3>
  <p>A simple calculator for product cost, selling price and profit.</p>
</div>

<div class="project-card">
  <h3>Student List</h3>
  <p>A webpage that displays students from a JavaScript array.</p>
</div>
</section>

<section id="contact">
  <h2>Contact Me</h2>
  <p>Email: abu@example.com</p>
</section>

```

An id should normally identify one specific element. A class can be reused on many elements. That is why both project boxes can use class="project-card".

8. Add the footer

```

<footer>
  <p>&copy; 2026 Abu Developer</p>
</footer>

```

© is an HTML entity that displays the copyright symbol.

9. Complete HTML

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Abu Portfolio</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>Abu Developer</h1>
    <nav>
      <a href="#about">About</a>
      <a href="#skills">Skills</a>
      <a href="#projects">Projects</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <main>
    <section class="hero">
      <h2>Hello, I'm Abu</h2>
      <p>I am learning web development and building useful websites.</p>
      <a class="button" href="#projects">View My Projects</a>
    </section>

    <section id="about">
      <h2>About Me</h2>
      <p>I am a beginner web developer learning HTML, CSS and JavaScript.</p>
    </section>

    <section id="skills">

```

```

    <h2>My Skills</h2>
    <ul>
      <li>HTML</li>
      <li>CSS</li>
      <li>JavaScript - Beginner</li>
    </ul>
  </section>

  <section id="projects">
    <h2>My Projects</h2>
    <div class="project-card">
      <h3>Shop Calculator</h3>
      <p>A simple calculator for product cost, selling price and profit.</p>
    </div>
    <div class="project-card">
      <h3>Student List</h3>
      <p>A webpage that displays students from a JavaScript array.</p>
    </div>
  </section>

  <section id="contact">
    <h2>Contact Me</h2>
    <p>Email: abu@example.com</p>
  </section>
</main>

<footer>
  <p>&copy; 2026 Abu Developer</p>
</footer>

</body>
</html>

```

10. Style the page with CSS

Now open style.css. We will style the HTML from the outside inward.

Reset and body

```

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  font-family: Arial, sans-serif;
  line-height: 1.6;
  background: #f4f4f4;
  color: #222;
}

```

The * selector targets every element. box-sizing: border-box makes width calculations easier. margin: 0 removes the browser's default outer spacing. line-height increases readability.

Header and navigation

```

header {
  background: #222;
  color: white;
  padding: 20px;
  text-align: center;
}

nav a {

```

```
color: white;
text-decoration: none;
margin: 0 10px;
}
```

The header gets a dark background and white text. text-decoration: none removes the default underline from navigation links. margin adds horizontal space between links.

Main content and sections

```
main {
  max-width: 900px;
  margin: auto;
}

section {
  background: white;
  margin: 20px;
  padding: 30px;
  border-radius: 10px;
}
```

max-width prevents the content from becoming too wide on large monitors. margin: auto centers the main content. Section padding creates space inside each white box, while margin creates space outside it.

Hero and button

```
.hero {
  text-align: center;
  padding: 60px 30px;
}

.button {
  display: inline-block;
  background: #222;
  color: white;
  padding: 10px 20px;
  text-decoration: none;
  border-radius: 5px;
}
```

The dot before hero and button means these are class selectors. display: inline-block lets the link behave more like a button while still remaining inline with surrounding content.

Project cards and footer

```
.project-card {
  border: 1px solid #ddd;
  padding: 15px;
  margin: 10px 0;
  border-radius: 8px;
}

footer {
  text-align: center;
  background: #222;
  color: white;
  padding: 15px;
}
```

Responsive CSS

```
@media (max-width: 600px) {
  nav a {
    display: block;
    margin: 8px 0;
  }
}
```

```

}

section {
  margin: 10px;
  padding: 20px;
}
}

```

A media query applies special CSS when a condition is true. Here, when the screen is 600 pixels wide or smaller, navigation links become vertical and sections use less spacing.

11. CSS selector cheat sheet

Selector	Meaning	Example
element	Targets every matching HTML tag	section { }
.class	Targets elements with a class	.project-card { }
#id	Targets an element with an ID	#about { }
element element	Targets an element inside another	nav a { }
*	Targets all elements	* { }

12. Understanding the CSS Box Model

Every HTML element can be imagined as a rectangular box with four layers:

- Content - the text, image or other content.
- Padding - space between the content and the border.
- Border - the line around the element.
- Margin - space outside the border, separating it from other elements.

```

section {
  margin: 20px;      /* outside space */
  padding: 30px;     /* inside space */
  border: 1px solid #ddd;
}

```

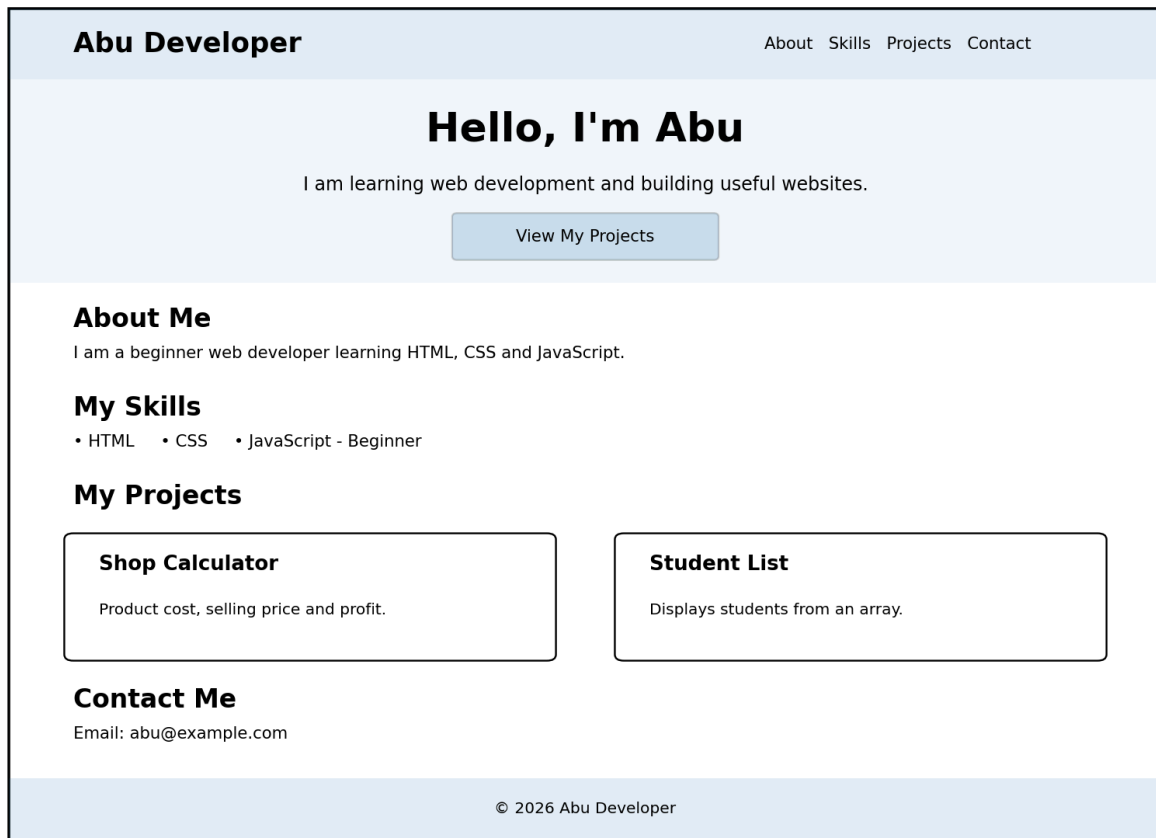
13. How to run the project

Save index.html and style.css. Open index.html in Chrome, Edge or Firefox. If you are using VS Code, you can install Live Server, right-click index.html, and choose Open with Live Server.

Debugging tip: If the page has HTML but no CSS design, first check that style.css is saved in the same folder and that the link tag says href="style.css".

14. Expected final result

Your finished webpage should have a dark header, centered introduction, white content sections, project cards and a dark footer. On a smaller screen, the navigation links should stack vertically.



15. Practice exercises

1. Change Abu to your own name.
2. Write your own About Me paragraph.
3. Add at least four skills.
4. Create a third project card.
5. Change the background and header colors.
6. Add your GitHub or LinkedIn link.
7. Add a profile image using the `<img>` tag.
8. Change the project-card border-radius and observe the difference.
9. Change the media query from 600px to 768px and test the browser width.

16. Student mini challenge

Build your own version without copying the finished code line by line. Your portfolio should contain:

- Your name and short introduction
- About section
- At least four skills
- At least three projects
- Contact information
- Navigation menu
- Responsive styling

17. What to learn next

Once you are comfortable with this portfolio, add JavaScript. Good beginner improvements are a dark-mode button, a contact form, a button click event, and dynamically displaying projects from a JavaScript array.